

BOLU, TurkeyWMO: **170700**Lat: **40.7338N**Lon: **31.6016E**Elev: **743**StdP: **92.71**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.4	-10.8	1.6	-7.9	-8.7	2.0	-5.1	4.4	6.2	4.0	4.6	1.0	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	31.8	19.6	30.0	19.5	28.4	19.0	21.7	28.5	20.8	27.4	19.9	26.4	2.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.6	25.7	18.5	14.7	24.4	17.7	13.9	23.5	67.2	28.6	63.7	27.5	60.5	26.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.3	3.7	3.3	-13.0	35.7	2.8	1.8	-15.0	37.0	-16.7	38.1	-18.3	39.1	-20.3	40.4		
(5)			-13.4	23.6	2.8	0.8	-15.4	24.2	-17.0	24.7	-18.5	25.1	-20.6	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	2.0	3.4	6.0	10.5	15.2	18.4	21.0	21.4	17.5	13.2	7.8	3.3
	DBStd	7.78	4.21	4.61	4.35	4.17	3.52	2.89	2.45	2.50	3.08	3.79	4.02	4.50
	HDD10.0	933	249	188	136	42	4	0	0	0	0	14	87	212
	HDD18.3	2687	507	418	382	236	105	33	5	6	51	162	317	466
	CDD10.0	1551	1	3	13	58	167	253	342	353	224	114	20	4
	CDD18.3	263	0	0	0	1	9	36	88	100	25	3	0	0
	CDH23.3	2579	0	0	2	29	161	431	808	852	248	47	0	0
	CDH26.7	791	0	0	0	3	30	123	269	295	62	8	0	0
Wind	WSAvg	1.2	1.1	1.2	1.4	1.4	1.3	1.3	1.4	1.4	1.3	1.0	1.0	1.0
Precipitation	PrecAvg	551	59	42	48	50	62	53	29	26	28	40	46	68
	PrecMax	717	130	121	98	91	174	156	95	78	79	78	114	156
	PrecMin	377	18	7	10	7	24	9	2	0	0	5	3	15
	PrecStd	81	29	26	23	17	27	31	23	22	20	20	28	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	18.0	21.7	26.7	29.6	32.4	34.6	34.2	31.1	27.5	21.1	17.6
		MCWB	8.0	9.6	11.2	14.5	16.9	19.3	20.1	19.1	17.9	16.8	12.8	10.0
	2%	DB	11.9	15.0	18.7	23.3	27.1	29.9	31.5	31.7	28.4	24.3	18.1	13.7
		MCWB	7.4	8.8	10.1	13.4	16.3	19.3	20.3	19.8	17.7	15.3	11.3	8.8
	5%	DB	10.1	12.5	16.1	20.9	25.0	27.8	29.6	29.8	26.3	21.9	15.9	11.6
		MCWB	6.4	7.5	9.0	12.3	15.9	18.6	20.1	19.5	17.4	14.6	10.6	7.9
	10%	DB	8.5	10.3	13.6	18.4	23.0	25.8	27.7	28.0	24.0	19.3	13.8	9.8
		MCWB	5.5	6.6	8.1	11.3	15.2	18.0	19.4	19.2	16.5	13.9	9.6	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	10.7	12.2	15.9	18.9	21.8	23.1	22.9	20.4	18.1	13.7	10.8
		MCDB	12.9	16.3	19.7	23.5	26.1	28.6	30.2	29.5	27.2	24.8	18.8	16.2
	2%	WB	7.7	9.0	10.5	14.2	17.6	20.3	22.0	21.6	19.1	16.4	12.2	9.2
		MCDB	11.3	13.7	17.2	21.8	24.7	27.3	28.8	28.3	25.3	22.0	16.5	12.8
	5%	WB	6.6	7.9	9.4	12.9	16.5	19.3	21.0	20.7	18.1	15.3	11.1	8.2
		MCDB	9.8	12.1	15.2	19.6	23.2	26.0	27.4	27.4	24.4	20.3	14.9	11.3
	10%	WB	5.6	6.7	8.3	11.7	15.6	18.3	20.1	19.8	17.1	14.3	10.0	7.0
		MCDB	8.2	10.3	13.1	17.5	21.7	24.6	26.3	26.2	22.8	18.5	13.3	9.7
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.6	9.9	11.4	12.4	12.7	12.8	12.9	12.5	10.9	9.8	7.2
		MCDBR	10.1	12.8	15.2	16.2	16.5	16.9	16.3	16.6	16.7	15.4	14.0	10.0
		MCWBR	6.2	7.3	7.8	7.7	7.5	7.9	7.2	6.7	7.3	7.3	7.7	6.0
	5% WB	MCDBR	9.3	11.6	13.8	14.4	14.6	15.0	14.1	13.5	14.5	13.5	11.8	8.6
		MCWBR	5.9	6.8	7.3	7.3	7.2	7.8	7.0	6.4	6.8	6.7	6.7	5.3
Clear-Sky Solar Irradiance	taub		0.314	0.340	0.384	0.419	0.397	0.380	0.380	0.381	0.367	0.355	0.325	0.314
	taud		2.370	2.306	2.218	2.158	2.294	2.375	2.386	2.376	2.412	2.427	2.466	2.429
	Ebn at Noon		838	865	860	853	881	894	890	880	871	840	823	809
	Edh at Noon		84	104	128	145	130	120	118	116	104	92	77	74
All-Sky Solar Radiation	RadAvg		1.61	2.50	3.64	4.84	5.93	6.73	7.10	6.30	4.78	3.25	2.21	1.50
	RadStd		0.21	0.38	0.35	0.52	0.54	0.63	0.32	0.40	0.38	0.27	0.28	0.20